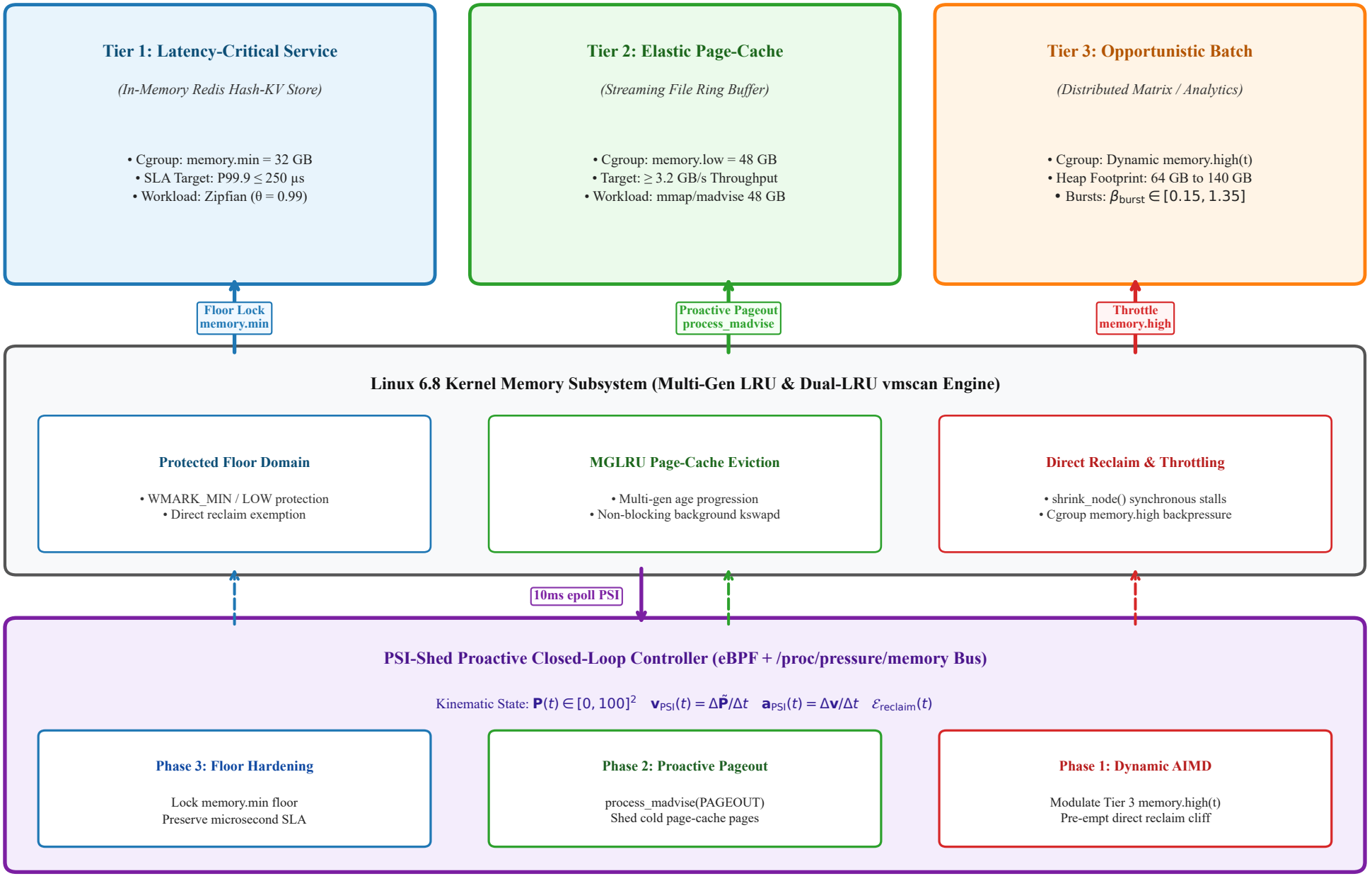

Tier 1: Latency-Critical Service
(In-Memory Redis Hash-KV Store)
• Cgroup: memory.min = 32 GB
• SLA Target: P99.9 ≤ 250 µs
• Workload: Zipfian (θ = 0.99)
Tier 2: Elastic Page-Cache
(Streaming File Ring Buffer)
• Cgroup: memory.low = 48 GB
• Target: ≥ 3.2 GB/s Throughput
• Workload: mmap/madvise 48 GB
Tier 3: Opportunistic Batch
(Distributed Matrix / Analytics)
• Cgroup: Dynamic memory.high(t)
• Heap Footprint: 64 GB to 140 GB
• Bursts: $\beta_{\text{burst}} \in [0.15, 1.35]$
Floor Lock memory.min
Proactive Pageout process_madvise
Throttle memory.high
Linux 6.8 Kernel Memory Subsystem (Multi-Gen LRU & Dual-LRU vmscan Engine)
Protected Floor Domain
• WMARK_MIN / LOW protection
• Direct reclaim exemption
MGLRU Page-Cache Eviction
• Multi-gen age progression
• Non-blocking background kswapd
Direct Reclaim & Throttling
• shrink_node() synchronous stalls
• Cgroup memory.high backpressure
10ms epoll PSI
PSI-Shed Proactive Closed-Loop Controller (eBPF + /proc/pressure/memory Bus)
Kinematic State: $\mathbf{P}(t) \in [0, 100]^2$ $\mathbf{v}_{\text{PSI}}(t) = \Delta\tilde{\mathbf{P}}/\Delta t$ $\mathbf{a}_{\text{PSI}}(t) = \Delta\mathbf{v}/\Delta t$ $\mathcal{E}_{\text{reclaim}}(t)$
Phase 3: Floor Hardening
Lock memory.min floor
Preserve microsecond SLA
Phase 2: Proactive Pageout
process_madvise(PAGEOUT)
Shed cold page-cache pages
Phase 1: Dynamic AIMD
Modulate Tier 3 memory.high(t)
Pre-empt direct reclaim cliff